Building Science

Lecture 3: Climate & Passive Environmental Control strategies

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Lecture outline

- Passive control strategies (continued)
- Thermal mass
- Natural ventilation

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Building Science

Course content

1. General introduction to Building Science
2. Climate impacts on Building orientation
3. Energy efficiency and Passive control strategies
4. Thermal performance – heat transfer principles
5. Thermal comfort
6. Thermal performance – Insulation and thermal bridges

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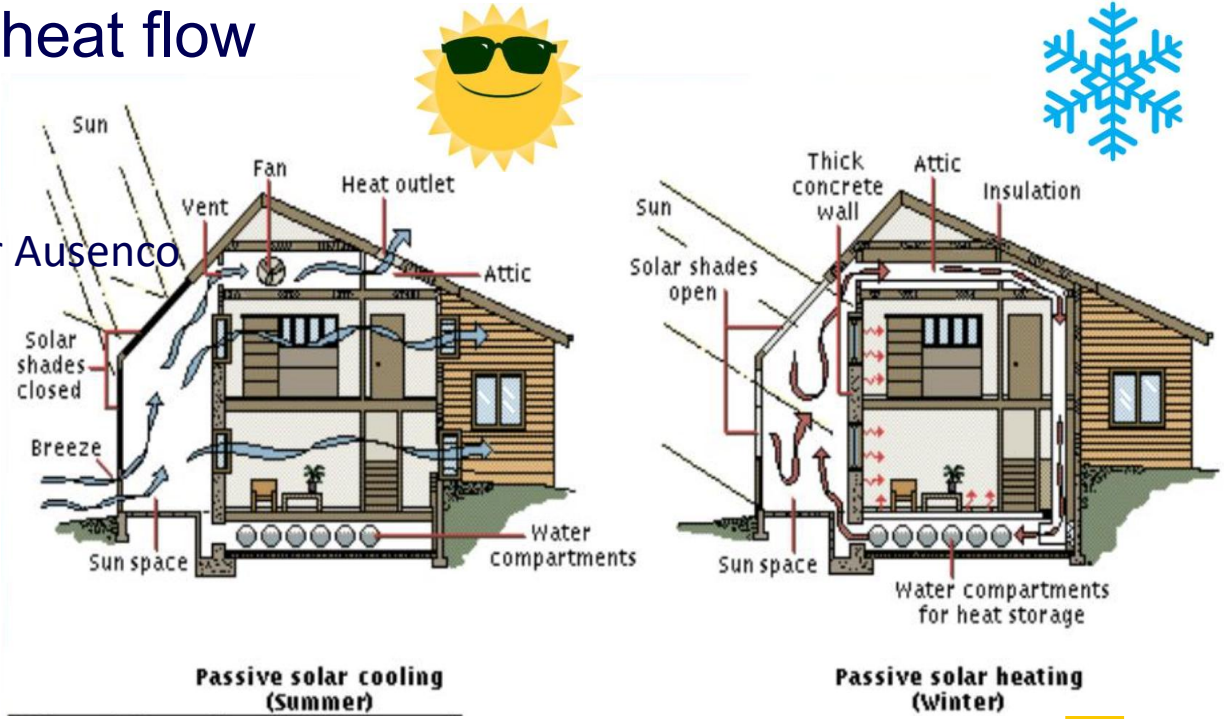
Climate strategies

Passive control of heat flow

- ○ Passive solar heating
- ○ Passive cooling
- ✓ Thermal mass
- ✓ Natural ventilation

Kramer Ausenco



Passive solar cooling (Summer)

Passive solar heating (Winter)

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Thermal mass

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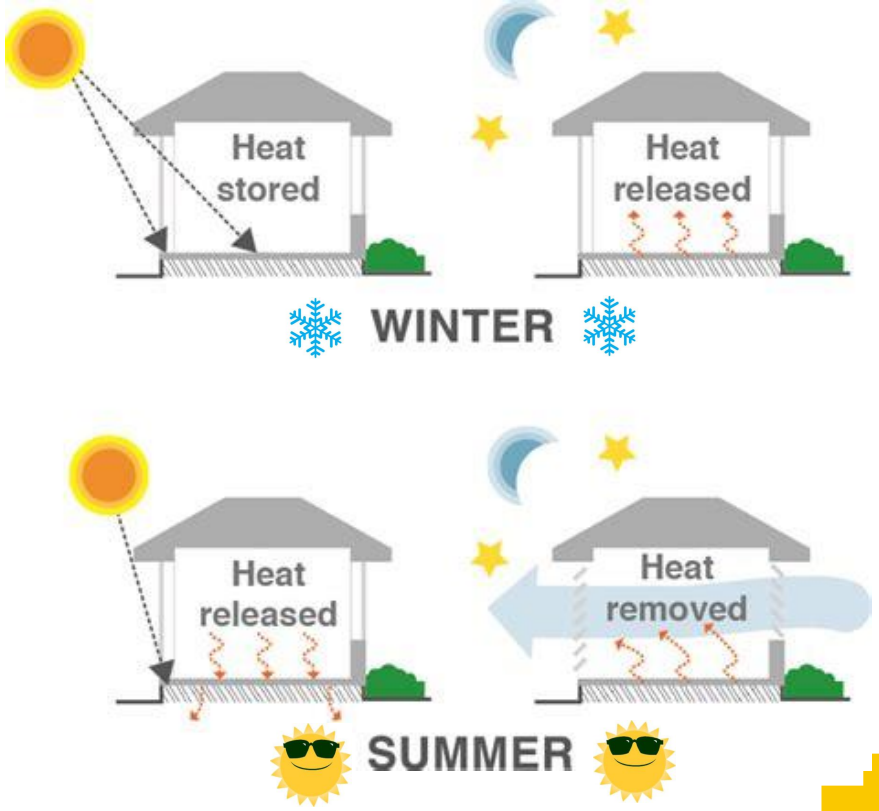
Climate strategies

Thermal mass

- ✓ Thermal mass **stores** heat as part of passive solar design
- ✓ Used to **delay** the passage of heat through the building fabric
- ✓ Used to **dampen** extreme internal temperatures



Climate strategies

Thermal mass

- ✓ Provided by heavy construction (e.g. thick walls, concrete, etc)
- ✓ Thick, dense walls take a long time to heat up
- ✓ This heat is dissipated overnight, keeping occupants warm
- ✓ In cold climates, it is beneficial for a continuously occupied building (e.g. home or hospital) but for an intermittently used and heated building (e.g. office or school), lightweight (well insulated) construction may be better
- ✓ In a hot-dry climate, with a large daily variation (diurnal variation), it may prevent the need for air-conditioning completely

Heavyweight construction	Lightweight construction
Concrete, steel, concrete block, other masonry	Timber, wood product

Natural ventilation

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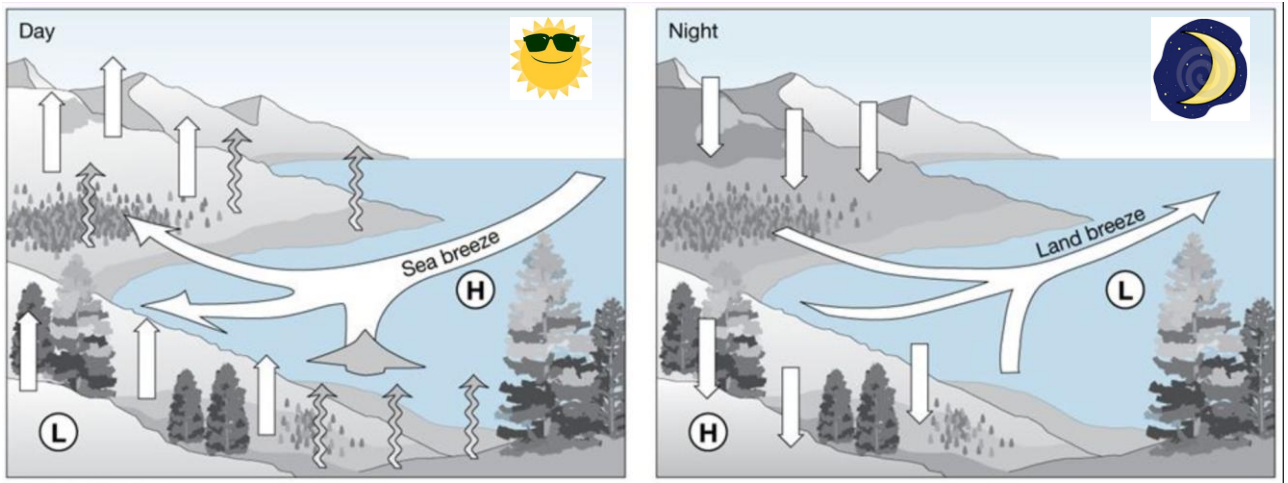
Climate strategies

Wind and ventilation

- ✓ Diurnal wind patterns
  - Onshore breeze during the day
  - Offshore breeze at night



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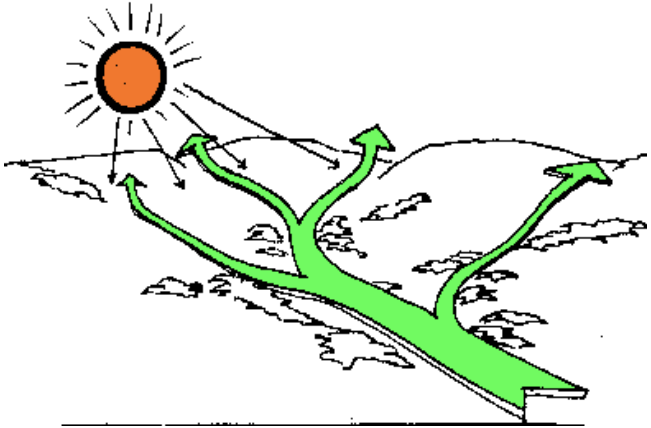
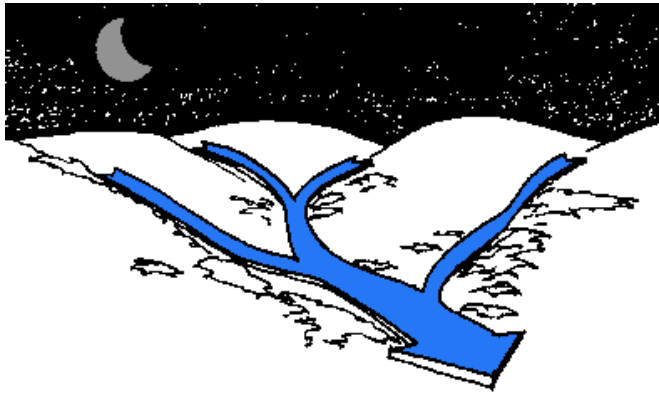
Climate strategies

Wind and ventilation

- ✓ Diurnal wind patterns
  - Up-valley as hillsides heat up
  - Down-valley as hillsides radiate to night sky
  - XuZhou !





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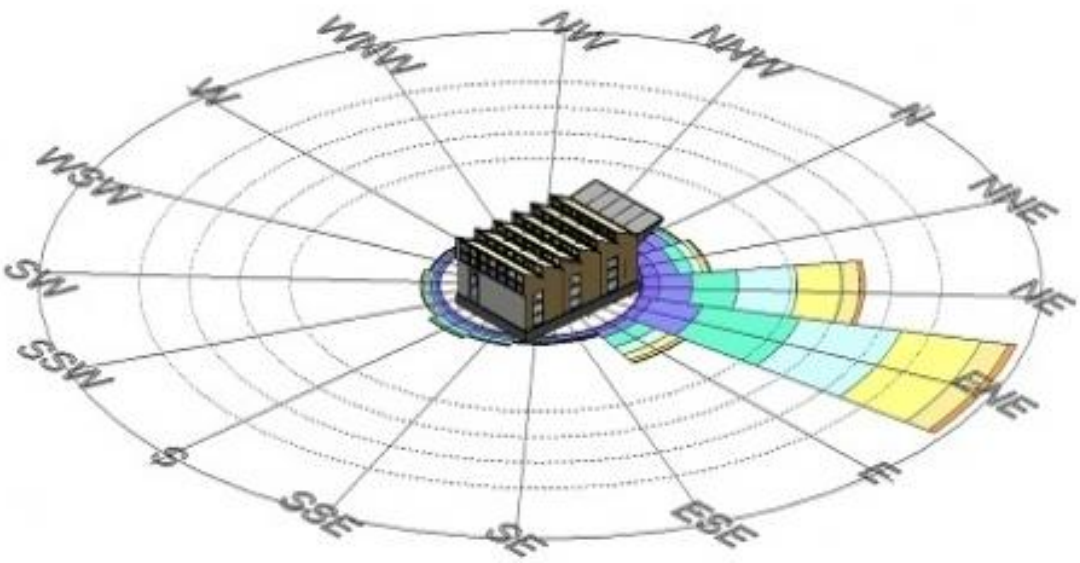
Climate strategies

Wind and ventilation

- ✓ Knowing your location's wind patterns influences important environmental and structural considerations for your building!



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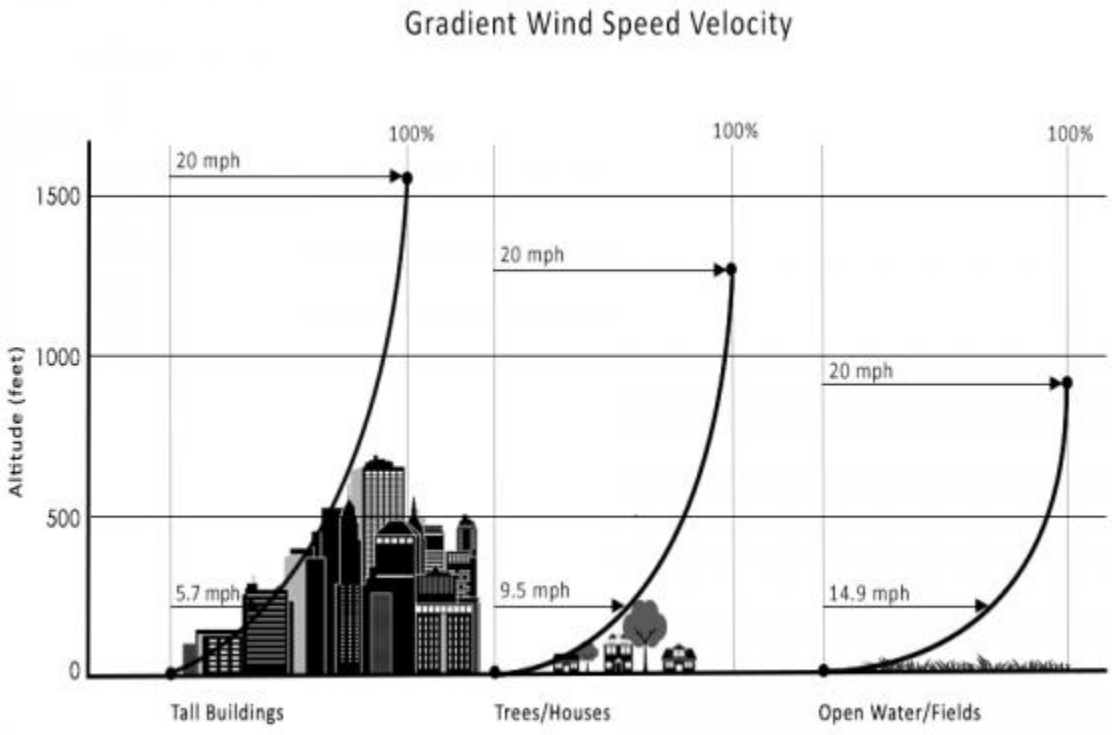
Climate strategies

Wind and ventilation

- ✓ Wind does not reach full speed until a certain height off the ground.
- ✓ This height depends on the local obstructions.



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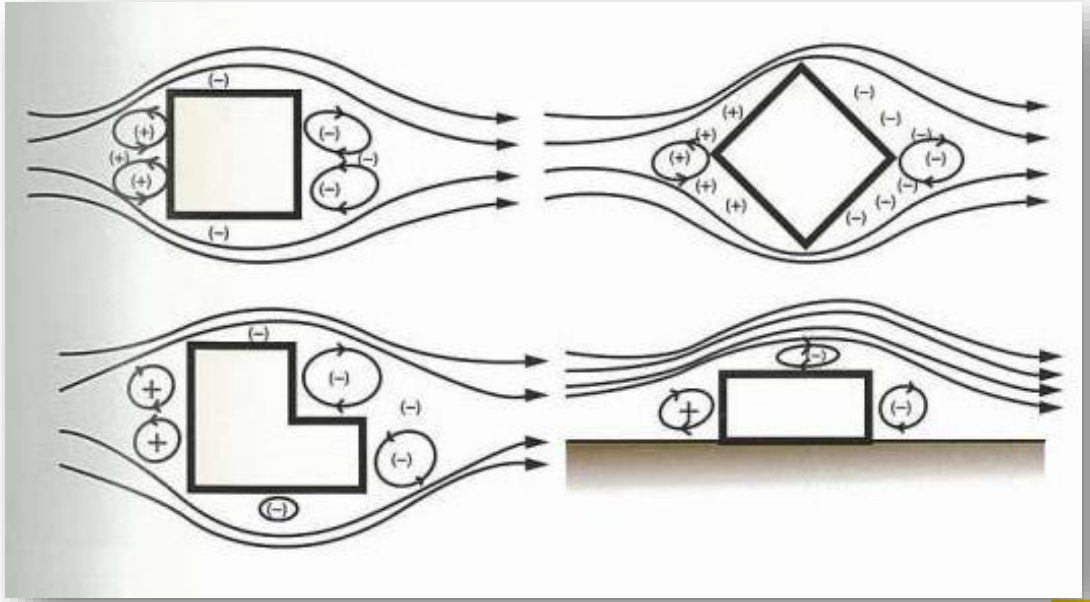
Natural ventilation strategies

Air movement principles

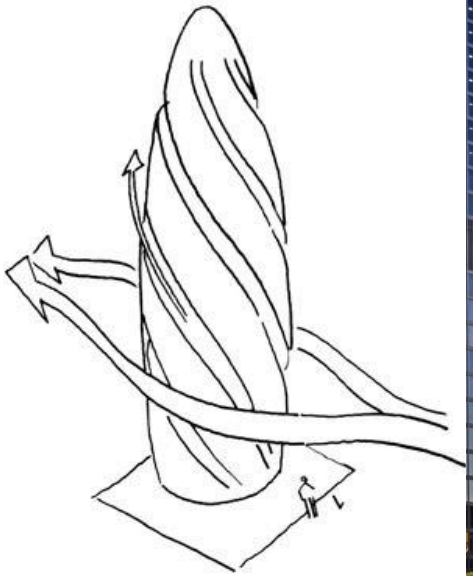
- ✓ Air tends to continue moving the same direction when it meets an obstruction

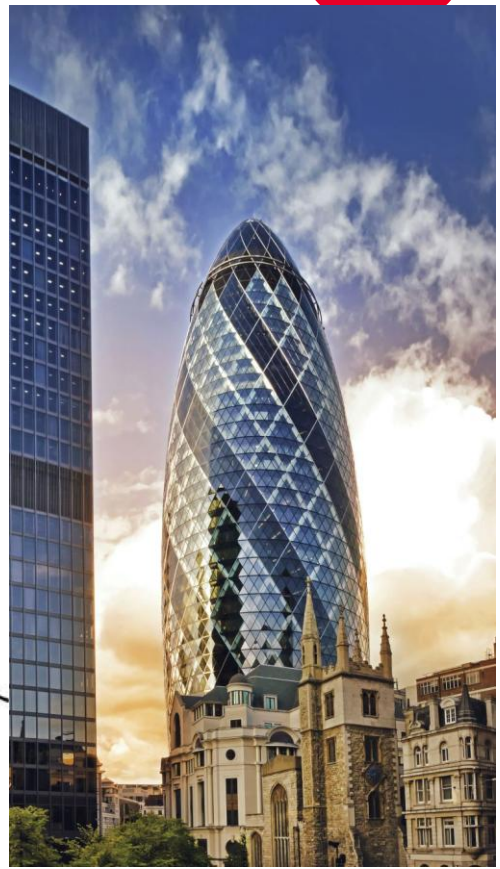


Natural ventilation strategies

Air movement principles

- ✓ The shape of building will greatly affect this





Source: Foster & Partners

Natural ventilation strategies

Air movement principles

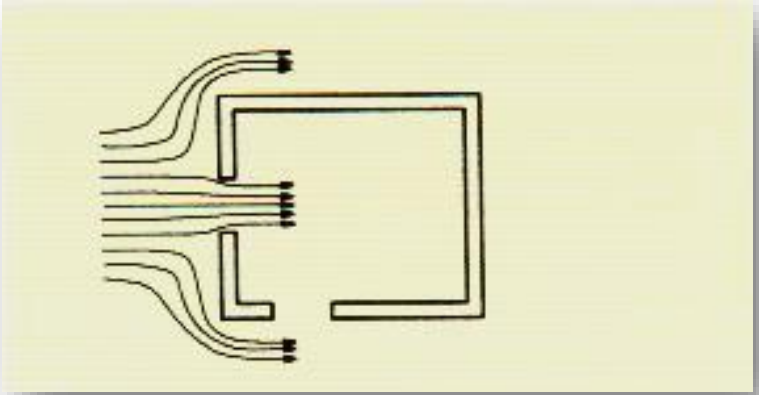
- ✓ Air flows from areas of **high pressure** to areas of **low pressure**

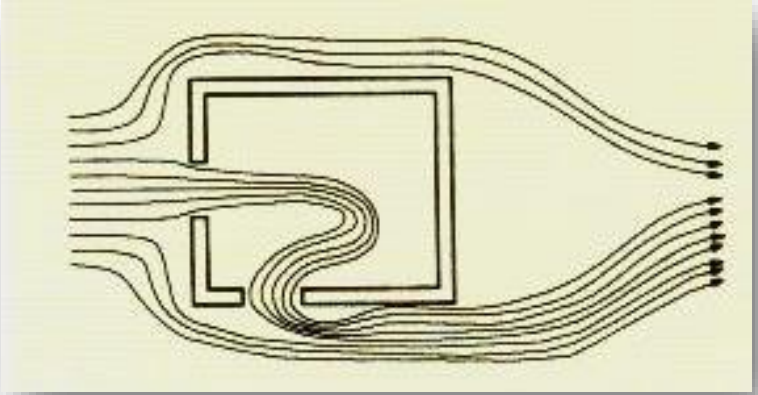
Wind Flow From High to Low Pressure Zones

Natural ventilation strategies

Air movement principles

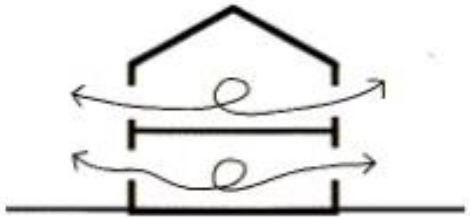
- ✓ One opening vs two openings (cross-ventilation)

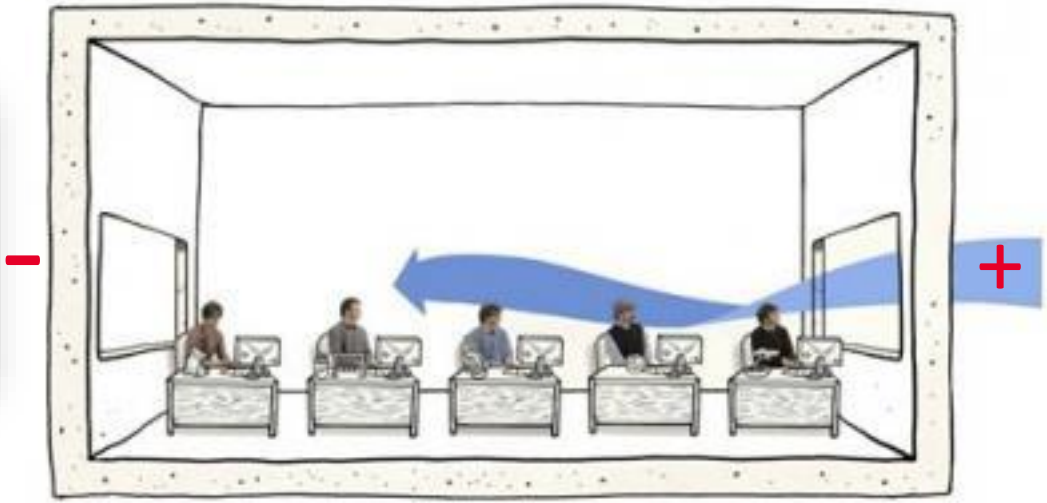




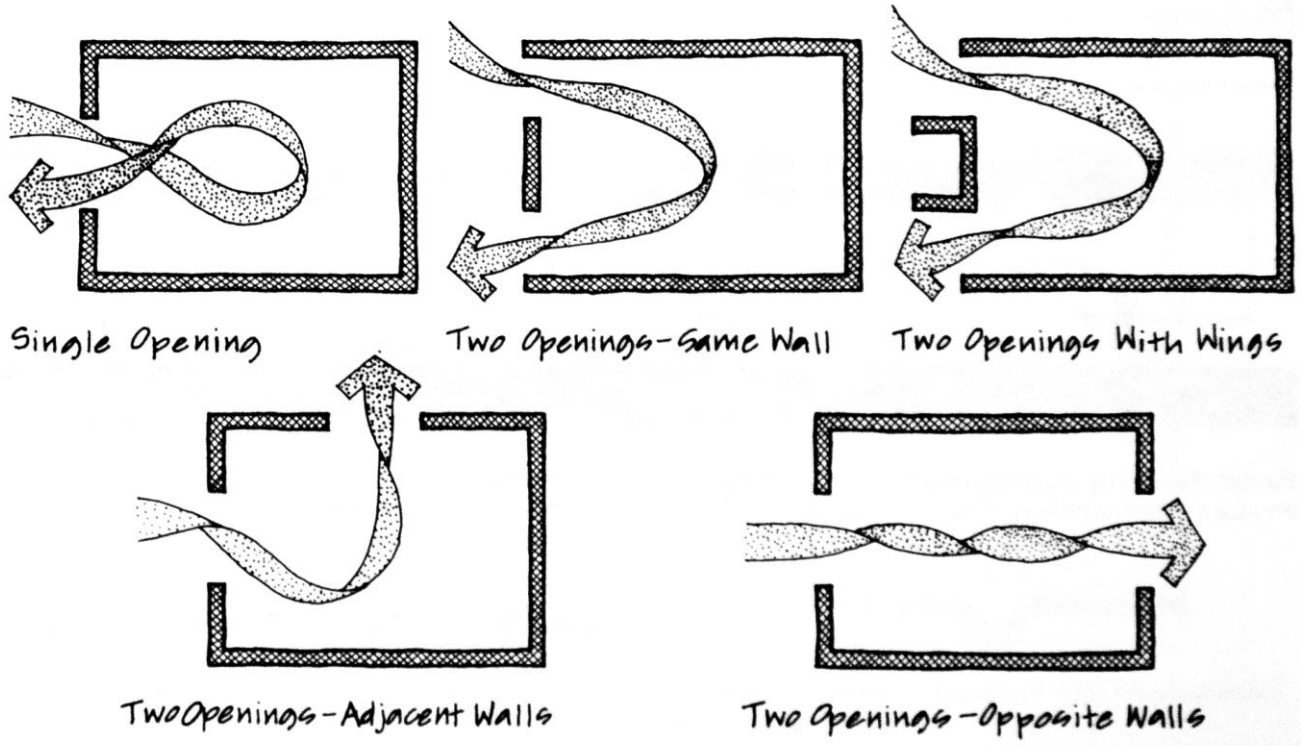
Natural ventilation strategies

Air movement is only possible if there is PRESSURE or TEMPERATURE differences between openings



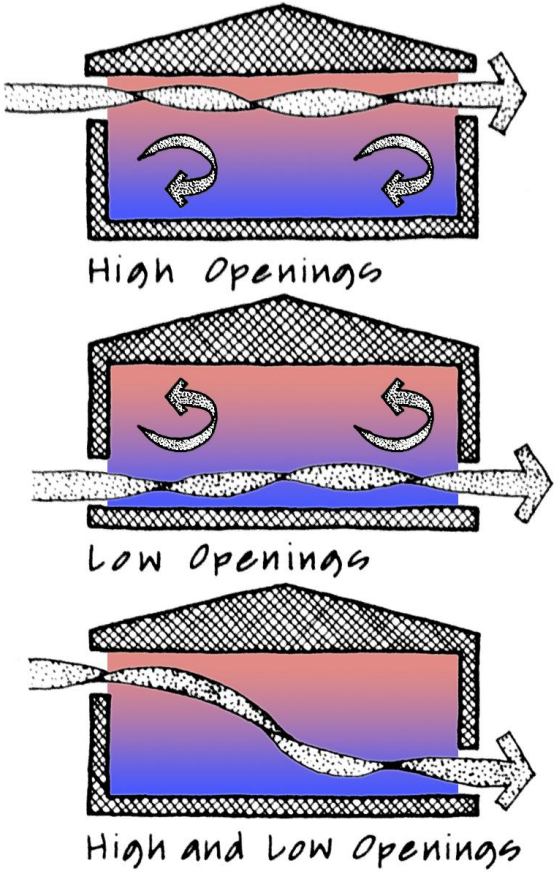


Natural ventilation strategies



Source: Sun, Wind, and Light, Brown & DeKay

Natural ventilation strategies



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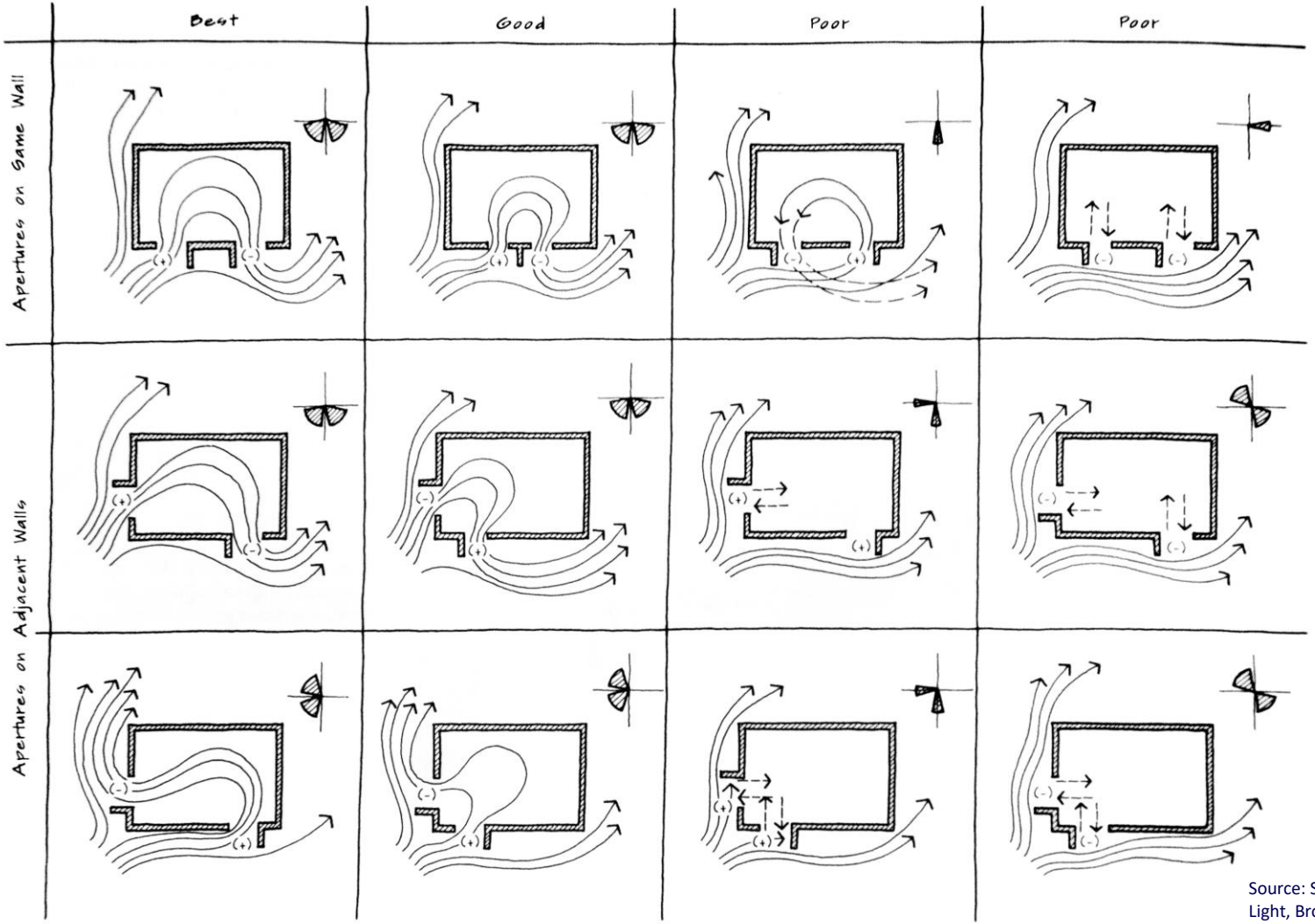
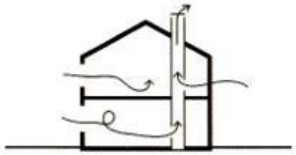


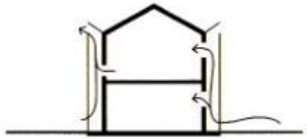
Source: Sun, Wind, and Light, Brown & DeKay

Natural ventilation strategies

Air movement by TEMPERATURE differences



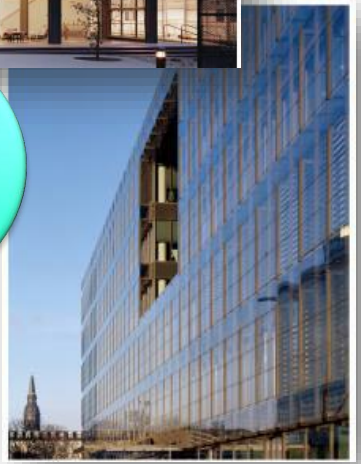
Chimneys and ducts



Double facades

Vertical spaces and atriums

'Stack Effect'



Natural ventilation strategies

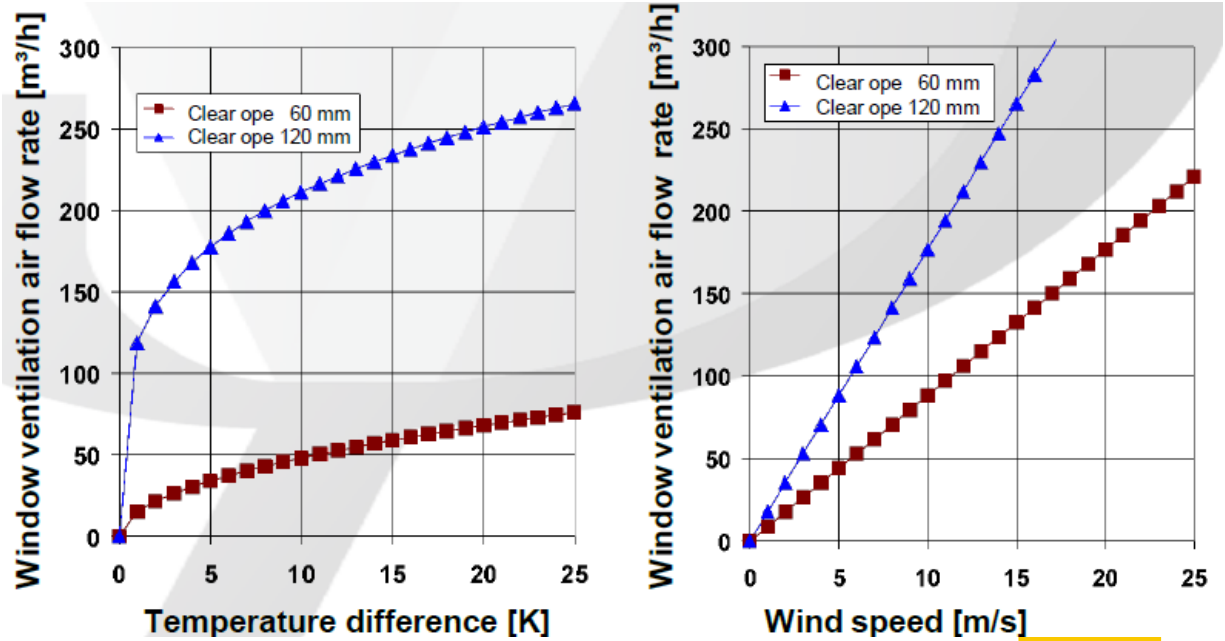
What might be some problems with natural ventilation?

- ✓ The incoming air cannot be filtered (e.g. dust, pollution, insects)
- ✓ The incoming air cannot be easily conditioned (e.g. temperature made cooler or hotter)
- ✓ The incoming air's humidity cannot be easily changed.

Natural ventilation strategies

Natural ventilation depends on external conditions!

- ✓ Temperature difference between inside and outside, and/or
- ✓ Wind speed



Natural ventilation strategies

Effective, consistent and weather-independent ventilation requires:

- ✓ A mechanical driving force (e.g. AHU fan) which assures constant air flow.
- ✓ Distributed air flow to each room in a building
- ✓ (Not guaranteed with only natural ventilation)

Thank you

Next time: Heat transfer principles

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